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Hiring Manager \ Evidence for Access, Technical Institute \ Genentech \ South San Francisco, CA

Dear Hiring Manager,

I am writing to apply for the Principal Data Scientist role on the Evidence for Access Technical Institute team. I want to be straightforward about fit: my recent work has been trial-level Bayesian modeling rather than large claims/EHR observational studies. I am applying because the methodological core of the role — designing defensible analyses that answer specific research questions from imperfect real-world data, with careful attention to confounding, censoring, and generalizability — is exactly what I do, and because the HEOR mission of turning evidence into access decisions is work I find genuinely meaningful.

At AstraZeneca, I build joint longitudinal–survival (“digital twin”) models of oncology tumor dynamics and multistate illness-death models for progression-free survival, used to forecast Phase 3 outcomes from Phase 2, historical trial, and real-world data. A significant part of that work is the real-world evidence side: designing weighted-likelihood schemes for borrowing from external cohorts, treating informative dropout and line-of-therapy switches as explicit model transitions rather than ignorable censoring, and being honest about what the RWD can and cannot tell us when trial and real-world event mixes differ. The statistical problems — selection, measurement, competing risks, heterogeneity across data sources — are the same problems observational researchers face; the tools differ, but transferring between propensity-score / regression-based designs and hierarchical Bayesian designs is a short step, not a long one, and one I expect to make quickly.

Three other pieces of my background are relevant. First, I hold a Ph.D. in Economics (Boston University, 2014) with fields in development, health, and experimental economics; my early-career work at Evidence Action designed and analyzed large-scale randomized evaluations in Kenya and Bangladesh, including an at-scale replication analysis whose null finding led to the program’s closure — a formative experience in caring about what the evidence actually says. Second, I have published in peer-reviewed economics journals (including a forthcoming paper in the *European Economic Review*) and have worked across academic, operational, and client-facing audiences. Third, at AstraZeneca I built and maintain an agentic-workflow layer on Claude Code — collaborative mathematical derivation, Stan diagnostic sub-agents, pipeline-management skills — shared across the modeling team through an internal plugin marketplace; I am comfortable both as an individual contributor on hard analytical problems and as someone who mentors and levels up team capability.

Where I would expect to ramp up: US claims and EMR data schemas, the US reimbursement coding system, and SAS/macro programming specifically. I am a strong programmer in R, Python, Stan, Julia, SQL, and C/C++, and I have worked productively on Databricks, Snowflake, and PostgreSQL backends, so the data-engineering side is familiar; the US-healthcare-specific vocabulary is what I would need to learn, and I am prepared to invest in that.

I would welcome the chance to discuss how the trial-modeling and RWE-integration work I have been doing could contribute to the E4A Technical Institute. Thank you for your consideration.

Sincerely,

Karim Naguib